Introduction to other file types

INTRODUCTION TO IMPORTING DATA IN PYTHON

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Other file types

- Excel spreadsheets
- MATLAB files
- SAS files
- Stata files
- HDF5 files

Pickled files

- File type native to Python
- Motivation: many datatypes for which it isn't obvious how to store them
- Pickled files are serialized
- Serialize = convert object to bytestream

Pickled files

```python
import pickle
with open('pickled_fruit.pkl', 'rb') as file:
    data = pickle.load(file)
print(data)
```

```
{'peaches': 13, 'apples': 4, 'oranges': 11}
```

Importing Excel spreadsheets

```
import pandas as pd
file = 'urbanpop.xlsx'
data = pd.ExcelFile(file)
print(data.sheet_names)
```

```
['1960-1966', '1967-1974', '1975-2011']
```

```
df1 = data.parse('1960-1966') # sheet name, as a string
df2 = data.parse(0) # sheet index, as a float
```

You'll learn:

- How to customize your import
- Skip rows
- Import certain columns
- Change column names

Let's practice!

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Importing SAS/Stata files using pandas

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SAS and Stata files

- SAS: Statistical Analysis System
- Stata: “Statistics” + “data”
- SAS: business analytics and biostatistics
- Stata: academic social sciences research

SAS files

- Used for:
  - Advanced analytics
  - Multivariate analysis
  - Business intelligence
  - Data management
  - Predictive analytics
  - Standard for computational analysis

Importing SAS files

```
import pandas as pd
from sas7bdat import SAS7BDAT
with SAS7BDAT('urbanpop.sas7bdat') as file:
    df_sas = file.to_data_frame()
```

Importing Stata files

```
import pandas as pd
data = pd.read_stata('urbanpop.dta')
```

Let's practice!

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Importing HDF5 files

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HDF5 files

- Hierarchical Data Format version 5
- Standard for storing large quantities of numerical data
- Datasets can be hundreds of gigabytes or terabytes
- HDF5 can scale to exabytes

Importing HDF5 files

```
import h5py
filename = 'H-H1_LOSC_4_V1-815411200-4096.hdf5'
data = h5py.File(filename, 'r') # 'r' is to read
print(type(data))
```

```
<class 'h5py._hl.files.File'>
```

The structure of HDF5 files

```
for key in data.keys():
    print(key)
```

```
meta
quality
strain
```

```
print(type(data['meta']))
```

```
<class 'h5py._hl.group.Group'>
```

This gives a high level picture of what's contained in a LIGO data file. There are 3 types of information:

- `meta`: Meta-data for the file. This is basic information such as the GPS times covered, which instrument, etc.
- `quality`: Refers to data quality. The main item here is a 1 Hz time series describing the data quality for each second of data. This is an important topic, and we'll devote a whole step of the tutorial to working with data quality information.
- `strain`: Strain data from the interferometer. In some sense, this is "the data", the main measurement performed by LIGO.

The structure of HDF5 files

```
for key in data['meta'].keys():
    print(key)
```

```
Description
DescriptionURL
Detector
Duration
GPSstart
Observatory
Type
UTCstart
```

```
print(np.array(data['meta']['Description']), np.array(data['meta']['Detector']))
```

```
b'Strain data time series from LIGO' b'H1'
```

The HDF Project

- Actively maintained by the HDF Group

- Based in Champaign, Illinois

Let's practice!

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Importing MATLAB files

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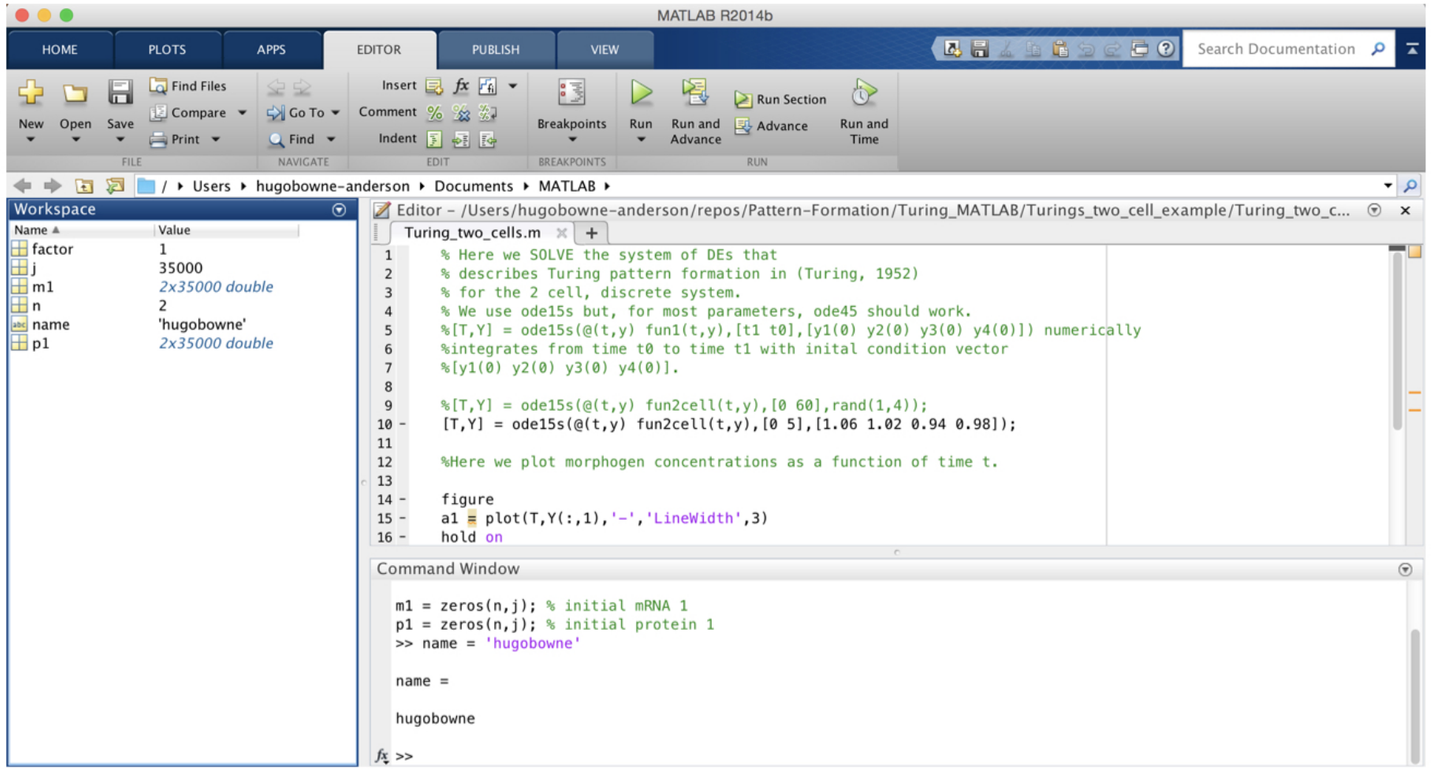
MATLAB

- “Matrix Laboratory”
- Industry standard in engineering and science
- Data saved as .mat files

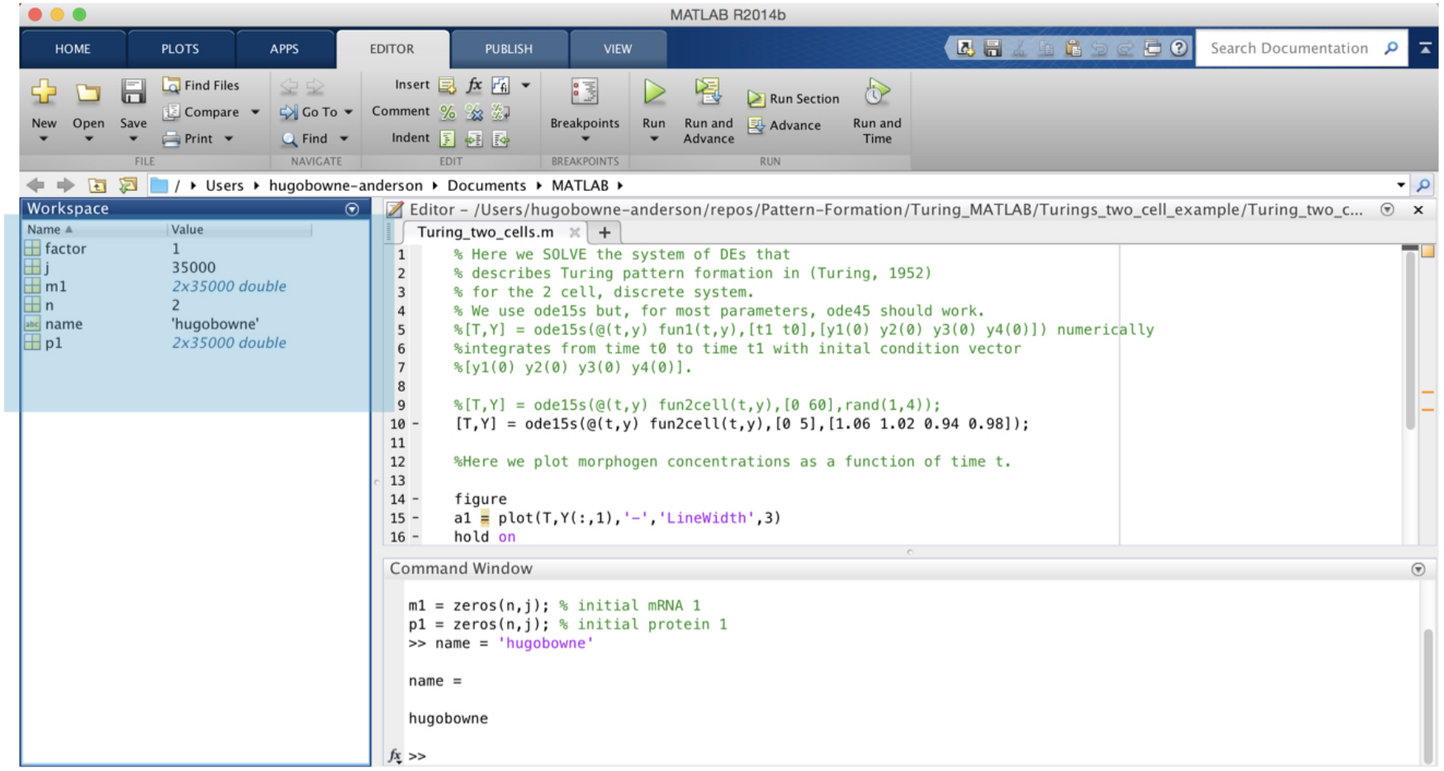
SciPy to the rescue!

- scipy.io.loadmat() - read .mat files
- scipy.io.savemat() - write .mat files

What is a .mat file?



What is a .mat file?



Importing a .mat file

```
import scipy.io
filename = 'workspace.mat'
mat = scipy.io.loadmat(filename)
print(type(mat))
```

```
<class 'dict'>
```

- keys = MATLAB variable names
- values = objects assigned to variables

```
print(type(mat['x']))
```

```
<class 'numpy.ndarray'>
```

Let's practice!

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